BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: 5191 / 5198

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 0922 (جماعت نہم) — سیکنڈری پارٹ I — Session 2018-20 to 2021-23 — ریاضی (معروضی) — گروپ پہلا

Q. Time: 20 minutes — Total Marks: 15 — PAPER CODE 5191

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. If |2 6; 3 x| = 0, then x = ___

(A) (B)

(C) (D)

Q2. y^(2/4) =

(A) (B)

(C) (D)

Q3. The logarithm of any number to itself as base is

(A) (B)

(C) (D)

Q4. If $x = 3+\sqrt{8}$, then $1/x =$

(A) (B)

(C) (D)

Q5. What will be added to complete the square of $9a^2-12ab$?

(A) (B)

(C) (D)

Q6. L.C.M. of a^2+b^2 and a^4-b^4 is

(A) (B)

(C) (D)

Q7. All the absolute value equations have ___ solutions.

(A) (B)

(C) (D)

Q8. If (x,0) = (0,y), then (x,y) is ___

(A) (B)

(C) (D)

Q9. Mid point of the points $(-3,0)$ and $(3,0)$ is

(A) (B)

(C) (D)

Q10. Number of right angles in a triangle can be

(A) (B)

(C) (D)

Q11. Medians of a triangle are ___

(A) (B)

(C) (D)

Q12. Symbol $\perp$ means.

(A) (B)

(C) (D)

Q13. Proportion is equality of ___ ratios.

(A) (B)

(C) (D)

Q14. Area of a parallelogram is equal to the ___ of base and height (altitude)

(A) (B)

(C) (D)

Q15. One angle on the base of an isosceles triangle is 30°. What is measure of its vertical angle.

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 0922 (**جماعت نہم**) — **سیکنڈری پارٹ** I — Session 2018-20 to 2021-23 — **ریاضی (انشائیہ)** — **گروپ پہلا** — **حصہ اول**

Q. (Marks 6×2=12) Time: 2:10 hours — Total Marks: 60 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے چھ اجزاء کے مختصر جوابات تحریر کریں۔ 6x2=12

(i) Check Matrix is singular or non-singular? D = [5 −10; −2 4] (ii) If A = [1 −2; 3 4], B = [0 7; −3 8] Find 3A−2B (iii) Simplify (8/125)^(−4/3) (iv) Simplify and write in a+ib fomn $(\sqrt{5}-3i)^2$ (v) Calculate $\log_3 2 \times \log_2 81$ (vi) Define Logarithm. (vii) Simplify $(\sqrt{2}+1/\sqrt{3})(\sqrt{2}-1/\sqrt{3})$ (viii) If $x=2-\sqrt{3}$ find 1/x (ix) Factorize $64x^3+27y^3$

Q. (Marks 6×2=12) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے چھ اجزاء کے مختصر جوابات تحریر کریں۔ 6x2=12

(i) Find H.C.F of the following polynomials. x^2+5x+6, $x^2-4x-12$ (ii) Solve the equation 3x/2 − (x−2)/3 = 25/6 (iii) Solve the equation $2\sqrt{(t+4)} = 5$ (iv) Find the value of m and c after expressing line 2x = y+3 in the form y = mx+c (v) Define Cartesian plane. (vi) Define coordinate geometry. (vii) Find the distance between the pairs of points. A(2,−6), B(3,−6) (viii) Define Parallelogram. (ix) The given figure LMNP is a parallelogram find the value of m,n. [Diagram: parallelogram LMNP with PN = 8, NM = 10, angle at L = 55°, angle at N = 55°, PL = 4m+n, LM = 8m−4n]

Q. (Marks 6×2=12) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے چھ اجزاء کے مختصر جوابات تحریر کریں۔ 6x2=12

(i) Define bisector of an angle. (ii) Verify that 3cm, 4cm and 5cm are lengths of the sides of a triangle. (iii) Define Congruent triangles. (iv) Find the value of x [Diagram: right triangle with hypotenuse 13 cm, base 5 cm, perpendicular x] (v) Verify that a=16cm, b=30cm and c=34cm are sides of a right triangle. (vi) Define interior of a triangle. (vii) Find the area

[Diagram: parallelogram with base 8 cm and height 4 cm] (viii) Define incentre of the triangle. (ix) Construct a △XYZ in which mZX = 6.4cm, mYZ = 2.4cm, m∠Y = 90°

Q. (Marks 8 × 3 = 24) Part II — Note: Attempt any Three Questions. Q.9 is compulsory

نوٹ: کوئی سے تین سوالات کے جوابات تحریر کریں۔ سوال نمبر 9 لازمی ہے

Q5. 5.a. Solve the system of linear equations by Cramer's rule: 4x+2y = 8; 3x−y = −1

b. Simplify ((a^p/a^q)^(p+q)) ((a^q/a^r)^(q+r)) ÷ 5(a^p·a^r)^(p−r)

Q6. 6.a. Use logarithm to find value of $\sqrt[3]{(0.7214\times20.37/60.8)}$

b. Find value of $m^2+n^2+p^2$ if m+n+p=10 and mn+np+mp=27

Q7. 7.a. If (x+2) is a factor of $3x^2-4kx-4k^2$, then find the value of k.

b. Simplify to the lowest form. $(x^2+x-6)/(x^2-x-6) \times (x^2-4)/(x^2-9)$

Q8. 8.a. Solve the Equation (1/2)(x−1/6) + 2/3 = 5/6 + (1/3)(1/2−3x)

b. Construct the triangle ABC and draw the perpendicular bisectors of the sides of this triangle. mAB = 4cm, mBC = 4.8cm, mCA = 3.6cm

Q9. 9. Prove that Any point equidistant from the end points of a line segment is on the right bisector of it. ------OR------ Prove that Parallelograms on equal bases and having the same (or equal) altitude are equal in area.

> Note: Board typos kept verbatim: 'fomn', 'Congurent'.

GROUP – II

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 0922 (**جماعت نہم**) — **سیکنڈری پارٹ** I — Session 2018-20 to 2021-23 — **ریاضی** (**معروضی**) — **گروپ دوسرا**

Q. Time: 20 minutes — Total Marks: 15 — PAPER CODE 5198

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of

the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. The value of 'x' from the equation $\sqrt{(2x-3)} - 7 = 0$ is

(A) (B)

(C) (D)

Q2. If $(x-1, y+1) = (0,0)$, then (x,y) is

(A) (B)

(C) (D)

Q3. The distance between the pair of points (0,0) and $(-4,-3)$ is ___

(A) (B)

(C) (D)

Q4. The symbol used for parallel is

(A) (B)

(C) (D)

Q5. A parallelogram has vertices.

(A) (B)

(C) (D)

Q6. The symbol used for the line AB is

(A) (B)

(C) (D)

Q7. Congruent figures have ___ area

(A) (B)

(C) (D)

Q8. The right bisectors of the three sides of a triangle are ___

(A) (B)

(C) (D)

Q9. $[\sqrt{2}\ 0;\ 0\ \sqrt{2}]$ is called ___ matrix

(A) (B)

(C) (D)

Q10. Write (4)^(4/3) with radical form

(A) (B)

(C) (D)

Q11. log e = ___, where e ≈ 2.718

(A) (B)

(C) (D)

Q12. If $x = 4-\sqrt{17}$ then the value of 1/x is

(A) (B)

(C) (D)

Q13. The unit of ratio is

(A) (B)

(C) (D)

Q14. The factors of x^2-5x+6 are

(A) (B)

(C) (D)

Q15. H.C.F of (x^2-5x+6) and (x^2-x-6) is

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 0922 (**جماعت نہم**) — **سیکنڈری پارٹ** I — Session 2018-20 to 2021-23 — **ریاضی (انشائیہ)** — **حصہ اول** — **گروپ دوسرا**

Q. (Marks 6×2=12) Time: 2:10 hours — Total Marks: 60 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے چھ اجزاء کے مختصر جوابات تحریر کریں۔ 6x2=12

(i) Define Scalar matrix. (ii) Find the product of [1 2] × [5; −4] (iii) Simplify using rules of exponents $((4a^3b^0)/(9a^{-5}))^{-2}$ (iv) Simplify and write the answer in a+bi: $(2-\sqrt{-4})(3-\sqrt{-4})$ (v) Find the value of x when $\log_{81}9 = x$ (vi) Calculate using laws of logarithm: $\log_5 3 \times \log_3 25$ (vii) Reduce the rational expression to the lowest form: $(120x^2y^3z^5)/(30x^3yz^2)$ (viii) Simplify $(3+\sqrt{3})(3-\sqrt{3})$ (ix) Factorize $x^2+x-132$

Q. (Marks 6×2=12) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے چھ اجزاء کے مختصر جوابات تحریر کریں۔ 6x2=12

(i) Use factorization to find the square root. $4x^2-12x+9$ (ii) Solve the equation $(x-3)/3-(x-2)/2=-1$ (iii) Solve the inequality. $3x+1<5x-4$ (iv) Find the value of m and c of the line $3x+y-1=0$ by expressing it in the form $y=mx+c$ (v) Verify whether the point (2,3) lies on the line $2x-y+1=0$ or not (vi) Define Equilateral Triangle. (vii) Find the distance between the pair of points. A(3,−11), B(3,−4) (viii) What is meant by congruency of Triangles? (ix) If the given figure ABCD is a parallelogram, then find x, m. [Diagram: parallelogram ABCD with angle at D = (5m+10)°, angle at C = 11x°, angle at A = 55°]

Q. (Marks 6×2=12) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے چھ اجزاء کے مختصر جوابات تحریر کریں۔ 6x2=12

(i) Define concurrent lines. (ii) 4cm, 7cm and 3cm are not the lengths of the triangle, Give the reason. (iii) When are two triangles, triangle ABC and triangle DEF called similar. (iv) Find the value of x [Diagram: right triangle with perpendicular 4 cm, base 3 cm, hypotenuse x] (v) Verify that a=1.5cm, b=2cm and c=2.5cm are sides of a right triangle. (vi) Define triangular region. (vii) Find the area [Diagram: square with side 4 cm] (viii) Define centroid of the triangle. (ix) Construct △ABC in which mAB = 3cm, mAC = 3.2cm, m∠A = 45°

Q. (Marks 8 × 3 = 24) Part II — Note: Attempt any Three Questions. Q.9 is compulsory

نوٹ: کوئی سے تین سوالات کے جوابات تحریر کریں۔ سوال نمبر 9 لازمی ہے

Q5. 5.a. Using Cramer's rule to solve Linear Equation. $3x-2y=1;\ -2x+3y=2$

b. Simplify $(a^{2l}/a^{l+m})(a^{2m}/a^{m+n})(a^{2n}/a^{n+l})$

Q6. 6.a. Use logarithm to find the value of $(8.97)^3\times(3.95)^2/\sqrt[3]{15.37}$

b. If $x+1/x=3$, find x^4+1/x^4

Q7. 7.a. Factorize by factor theorem $x^3-x^2-22x+40$

b. Use division method to find the square root. $4x^2+12xy+9y^2+16x+24y+16$

Q8. 8.a. Solve the inequality $3x-2<2x+1<4x+17$

b. Construct triangle PQR. Draw its altitudes. mPQ = 4.5cm, mQR = 3.9cm and m∠R = 45°

Q9. 9. Prove that "Any point on the bisector of an angle is equidistant from its arms." ------OR------ Prove that "Triangles on the same base and of same altitude are equal in area"

> Note: Q4(vii) diagram is a square (4 cm side) though the question says 'Find the area'. Board typos kept verbatim: 'followings', 'Congurent'.